

Project File

Name :- Anupma Mishra

Reg no. :- 25BAI11311

Branch :- B.tech in CSE AI & ML

Course Code :- CSE2006

Course Title :- Programming in Java

Project Type:- Console-Based Application

Date :- 18/09/2026

University :- VIT Bhopal University

COLLEGE EVENT MANAGEMENT SYSTEM

1. INTRODUCTION

The College Event Management System is a console-based Java application developed to simplify the management of college events and student registrations. The system provides separate functionalities for students and administrators.

Students can view, search, register for, and cancel events, while administrators can add, update, and delete events. The system also manages seat availability and prevents duplicate registrations.

The project demonstrates the practical implementation of Object-Oriented Programming, Collections, Exception Handling, and File Handling in Java.

2. PROBLEM STATEMENT

Managing college events manually can lead to difficulties in maintaining event information, tracking student registrations, monitoring available seats, and avoiding duplicate registrations.

The proposed system provides a simple computerized solution for managing these activities through a menu-driven Java application.

3. OBJECTIVES

The main objectives of the project are:

- * To develop a simple college event management application.
- * To allow students to view and search events.
- * To provide event registration and cancellation facilities.
- * To allow administrators to manage event details.
- * To automatically update available seats.
- * To prevent duplicate registrations.
- * To implement exception handling for invalid situations.
- * To demonstrate Java Object-Oriented Programming concepts.
- * To implement file handling for storing event and registration data.

4. SCOPE OF THE PROJECT

The system focuses on basic college event management activities.

Student

- * View events
- * Search events
- * Register for events
- * Cancel registrations
- * View registered events

Administrator

- * View events
- * Add events
- * Update events
- * Delete events

The system is designed as a console-based application and can be extended into a GUI or web application in the future.

5. TECHNOLOGIES USED

Technology	Purpose
Java	Application development
ArrayList	Managing events and registrations
OOP	Program structure and design
Exception Handling	Handling errors
File I/O	Saving data to text files
Visual Studio Code	Development environment
Git & GitHub	Version control and project submission

Note: JDBC, MySQL, JPA, ORM, and JPQL are not used in the current implementation.

6. SYSTEM FEATURES

Event Management

Administrators can create, update, view, and delete events.

Each event contains:

- * Event ID

- * Event Name

- * Category

- * Date

- * Time

- * Venue

- * Organizer

- * Total Seats

- * Available Seats

Registration Management

Students can register for available events. The system checks whether the event exists, whether seats are available, and whether the student has already registered.

Seat Management

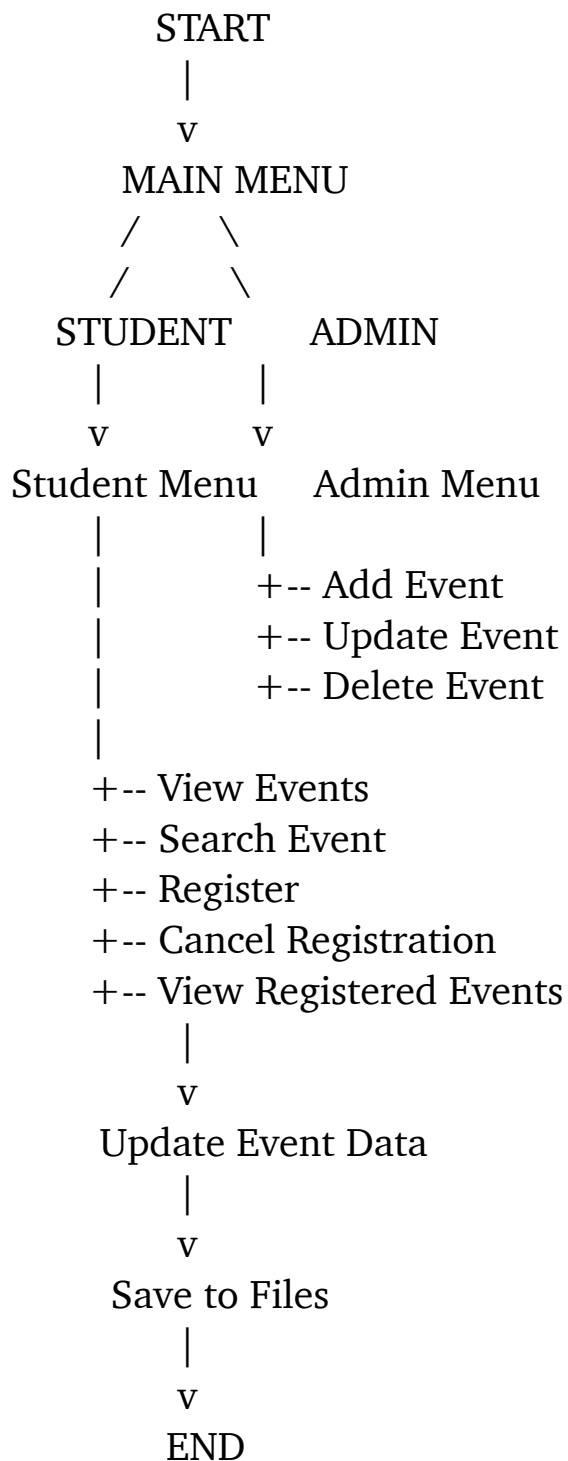
When a student registers:

Available Seats = Available Seats - 1

When a student cancels:

Available Seats = Available Seats + 1

7. SYSTEM WORKFLOW



8. CLASS DESCRIPTION

Main.java

Contains the main program, menus, event management operations, registration operations, and input validation.

Event.java

Represents an event and stores event-related information such as name, date, venue, organizer, and seat availability.

User.java

An abstract class containing common user information and the abstract showRole() method.

Student.java

Extends the User class and represents a student.

Admin.java

Extends the User class and represents an administrator.

Registration.java

Stores the relationship between a student and an event.

FileManager.java

Handles saving event and registration information into text files.

EventNotFoundException.java

Custom exception used when an event cannot be found.

EventFullException.java

Custom exception used when an event has no available seats.

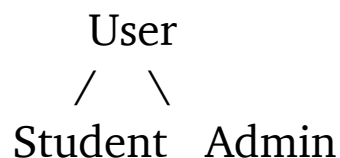
9. OBJECT-ORIENTED PROGRAMMING CONCEPTS

Encapsulation

Private data members and getter/setter methods are used to control access to class data.

Inheritance

Student and Admin inherit common properties and methods from the User class.



Abstraction

User is an abstract class containing the abstract method showRole().

Polymorphism

The showRole() method is overridden by both Student and Admin.

10. EXCEPTION HANDLING

The project uses custom exceptions to handle specific errors.

EventNotFoundException

Generated when the entered Event ID does not exist.

EventFullException

Generated when an event has no available seats.

Exception handling prevents the application from terminating unexpectedly and provides meaningful error messages to the user.

11. FILE HANDLING

The project uses Java File I/O to save current event and registration information.

Two text files are used:

events.txt

registrations.txt

The FileManager class uses FileWriter to write the data into these files.

12. INPUT VALIDATION

The system validates numeric inputs to prevent invalid menu selections and seat values.

For example, if a user enters text instead of a number, the program displays an error message and requests valid numeric input.

The system also ensures that the number of total seats is greater than zero.

13. TESTING

The following functionalities were tested:

Test Case	Expected Result	Status
View Events	Events displayed	Passed
Search Valid Event	Event displayed	Passed
Search Invalid Event	Error message displayed	Passed
Add Event	Event added	Passed
Update Event	Event updated	Passed
Delete Event	Event deleted	Passed
Register Event	Registration successful	Passed
Duplicate Registration	Registration prevented	Passed
Cancel Registration	Registration cancelled	Passed
Seat Update	Seat count updated	Passed
Full Event	Exception handled	Passed
Invalid Input	Error handled	Passed

14. LIMITATIONS

- * The application uses a console-based interface.
- * No database is used in the current implementation.
- * Data is saved in text files.
- * Automatic loading of saved data when the application starts is not implemented.
- * User authentication is not implemented.
- * Multiple users cannot access the system simultaneously.

15. FUTURE SCOPE

The project can be enhanced by implementing:

- * Database integration
- * JDBC connectivity
- * GUI or web interface
- * User authentication
- * Automatic data loading
- * Email notifications
- * Event reminders
- * Event filtering and advanced search
- * Reports and analytics

These features are considered future enhancements and are not part of the current implementation.

16. CONCLUSION

The College Event Management System successfully demonstrates the development of a menu-driven Java application for managing college events and student registrations.

The project implements important Java concepts including Object-Oriented Programming, inheritance, abstraction, polymorphism, collections, exception handling, input validation, and file handling.

The system provides a simple and organized way to manage events, registrations, and seat availability while providing a foundation for future development into a database-driven or web-based application.

17. REFERENCES

1. Java Programming concepts and Object-Oriented Programming principles.
2. Java Collections Framework.
3. Java Exception Handling.
4. Java File I/O.
5. Git and GitHub documentation.